

Group A

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Exp #7: Dipole Moment of Polar Molecules in Solution

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1. Measurement of Dielectric Constants

x 1,2-dichlorobenzene

(1) 0.010 M 2.10

(2) 0.020 M 2.17

(3) 0.030 M 2.23

(4) 0.040 M 2.29

2. Measurement of Refraction

(1) 0.010 M : 1.4280 19.4°C

(2) 0.020 M : 1.4292 19.6°C

(3) 0.030 M : 1.4306 19.7°C

(4) 0.040 M : 1.4318 19.8°C